

Figure 1.29 Fixed Resistor

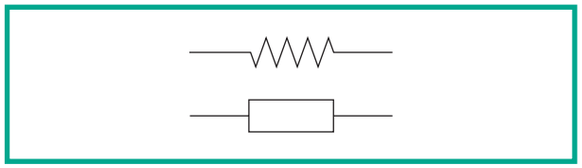


Figure 1.30 Symbols of fixed resistor

The classification of resistors depends upon their materialsand fabrication methods used.

1. Carbon resistors
2. Wire wound resistors
3. Metal oxide resistors
4. Metal film resistors
5. Printed resistors
6. SMD type (Surface Mount Device).

In modern circuits, Printed resistors and SMD type resistors are used.

Special Type of Resistors

7. Positive Temperature Coefficient Resistor (PTC)
8. Negative Temperature Coefficient Resistor (NTC)

1.8.2 Carbon Resistor

It is made up of carbon paste and burnt clay with connecting leads, which is heated and kept under high pressure. The resistor is formed as small sticks. It can handle small amount of current through it. If high current flows through this type of resistor, it will burn. Figure 1.31 shows the internal construction of a carbon resistor.

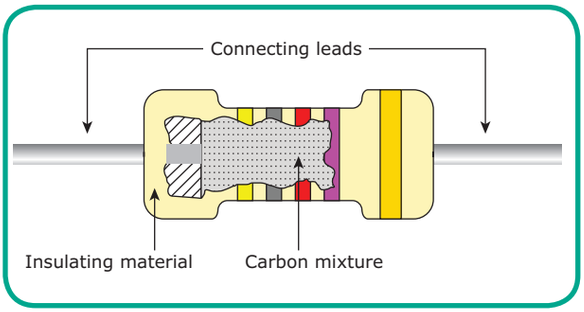


Figure 1.31 Carbon resistor

Uses

1. It is widely used in low current flowing circuits like Radio, DVD players and television etc.

1.8.3 Wire Wound Resistors

Figure 1.32 shows the cut away view of a wire wound resistor. Particular length of Nichrome is wound on a small Porcelain cylinder. The two terminals of the nichrome are connected to the copper legs. The surface is given a protective coating with glass and enamel. These types of resistors are manufactured with high precision.

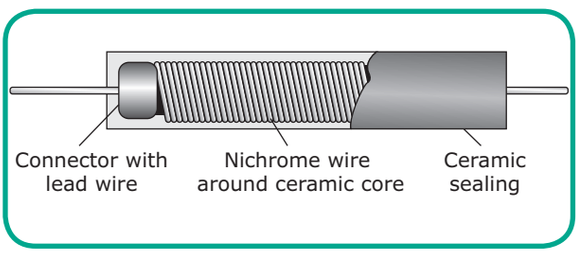


Figure 1.32 Wire wound resistor

Uses

1. Regulated power supplies in television, inverters etc.
2. Also used in Public Address system amplifiers.



Nichrome is an alloy made of Nickel and chromium.

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